

Data versus Information Sales under Financial Constraints

Dmitry Orlov (UW-Madison)

Andrzej Skrzypacz (Stanford)

Pavel Zryumov (Rochester)

SFS Cavalcade

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Data and Information Driven Economy

- ▶ Firms often have data that can benefit other market participants.
 - Example 1: viewer preference data generated by streaming platforms.
 - Example 2: driving data generated by Uber, Waymo, Tesla used to train self-driving algorithms.
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 - Its quality cannot be verified by a third party, e.g., court
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- ▶ To what extent can firms produce, share, and sell data efficiently?
- ▶ Many papers focus on the non-rival properties of information.
 - Jones and Tonetti (2020), Farboodi and Veldkamp (2025).
- ▶ This paper studies the implications of the uncertainty of information content for information trade.

Preview of the Paper

- ▶ Study information sharing without dynamic commitment.
 - Seller cannot commit how much information to share in the future.
 - Buyer cannot commit to future payments.
- ▶ Information is special: its content is inherently uncertain and the resolution of this uncertainty poses risks for efficient communication.

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- ▶ Both **dynamic communication** and **information design** are needed to achieve efficiency.
 - Uncertainty in signals implies that information and data may behave differently from capital goods.

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- ▶ Information is special: its content is inherently uncertain and the resolution of this uncertainty poses risks for efficient communication.
- ▶ Both **dynamic communication** and **information design** are needed to achieve efficiency.
 - Uncertainty in signals implies that information and data may behave differently from capital goods.
- ▶ Sharing granular signals exposes parties to inefficient risk due to the short-term uncertainty in data content.
 - Efficiency can never be achieved for discrete fundamentals.
 - Slack in the budget is necessary for any non-Gaussian fundamentals.

Related Literature

- ▶ **Gradualism:** Che and Sàcovic (2004), Compte and Jehiel (2004), Horner and Skrzypacz (2016), Pitchford and Snyder (1994), Lockwood and Thomas (2002), Admati and Perry (1991).
- ▶ **Dynamic Bayesian Persuasion:** Ely (2017), Ely and Szydlowski (2020), Orlov, Skrzypacz and Zryumov (2020), Brocas and Carillo (2007).
- ▶ **Multiple Receivers:** Rosenberg, Solan, and Vieille (2013), Kolotilin, Mylovanov, Zepechelnyuk and Li (2017), Orlov, Zryumov and Skrzypacz (2023), Inostroza (2019).

Model

Players and Preferences

- ▶ Time is continuous $t \in [0, 1]$ (keeps track of sequence of events).
- ▶ Fundamental **state** $\theta \in \{0, 1\}$, with common prior $p_0 = P(\theta = 1)$.
- ▶ Two players: Sender and Receiver.
- ▶ Quasi-linear preferences over **terminal** beliefs p and money m :

$$U_i(p, m) = u_i(p) + m \quad i \in \{R, S\}.$$

- Receiver likes information sharing: $u_R(p)$ is convex.
 - Sender dislikes information sharing: $u_S(p)$ is concave, e.g., competition.
- ▶ Gains from information (“trade”): $u_R(p) + u_S(p)$ is convex in p .
 - Full information sharing is efficient.
- ▶ Receiver has a starting budget $\bar{M}$, Sender is penniless.
 - Can the Receiver pay to induce efficient information sharing?

Information Flow and Timing

- ▶ Sender has access to a Gaussian d.g.p. $X = (X_i)_{i=0}^{\infty}$ about θ :

$$dX_i = \theta di + \sigma dB_i.$$

- ▶ Each “**data**” unit $d\mathbf{X}_i$ is an **infinitesimal granular signal** about θ .
 - Observing the entire data-stream fully reveals θ .
 - Infinitesimal signals are divisible $\Rightarrow$ no built-in need to split signals.

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- ▶ D_t : quantity of data shared by Sender up to time t .
 - Sender discloses process X to Receiver over time.
 - Both players observe X_i for $i \in [0, D_t]$.
- ▶ M_t : cumulative payment to the Sender up to time t .
 - Receiver makes payments to Sender over time.
 - Payments are irreversible, e.g., capital investments.

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Definition

Data-sharing strategy is a weakly increasing **continuous** process $D = (D_t)_{t \in [0,1]}$ that is **predictable** w.r.t. the induced filtration $\mathbb{F}$.

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- ▶ **Continuity**: Sender observes all “intermediate” data points as they are shared.
 - Cannot package data into blocks.

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- ▶ The “size” of the block can depend on the data contained in the block itself.
 - Intuitively, the Sender can buffer new data. Once the data in the buffer reaches some threshold, the Sender shares the entire buffer as a single block.
 - E.g. $D_t = \inf\{I > D_{t-} : X_I - X_{D_{t-}} > c\}$ for some $c > 0$.

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- ▶ Lemma: Information-sharing strategies allow for Bayesian Persuasion within period dt .

Hold-up Problem

- ▶ **Receiver Hold-up:** if Sender shares all data/information upfront, then Receiver can withhold payments.
- ▶ **Sender Hold-up:** if Receiver pays up-front, then Sender can withhold information.

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- ▶ **Sender Hold-up:** if Receiver pays up-front, then Sender can withhold information.
- ▶ Hold-up problems prohibit efficient communication with static strategies.
 - Even though the optimal efficient outcome is static: communicate all data/information in exchange for individually rational payment.
 - Even though they can design information flexibly, they cannot contract/commit to it.
 - Applies if data is not verifiable by a third party.

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 - Even though the optimal efficient outcome is static: communicate all data/information in exchange for individually rational payment.
 - Even though they can design information flexibly, they cannot contract/commit to it.
 - Applies if data is not verifiable by a third party.
- ▶ **Unique Static Nash** equilibrium = no data sharing and no payments.
 - Static Nash is the worst continuation equilibrium, e.g., punishment.
- ▶ Can the players do better if they communicated over several periods?

Equilibrium Definition

Definition

An equilibrium is a pair of the Sender's sharing strategy $D^* = (D_t^*)_{t \in [0,1]}$ and Receiver's payment strategy $M^* = (M_t^*)_{t \in [0,1]}$, that satisfy

- ▶ Sender incentive compatibility:

$$\mathbb{E} \left[\underbrace{u_S(p_1^*) + M_1^*}_{\text{Sender equilibrium payoff}} \mid \mathcal{F}_{t-} \right] \geq \underbrace{u_S(p_{t-}^*) + M_{t-}^*}_{\text{Sender deviation payoff}}$$

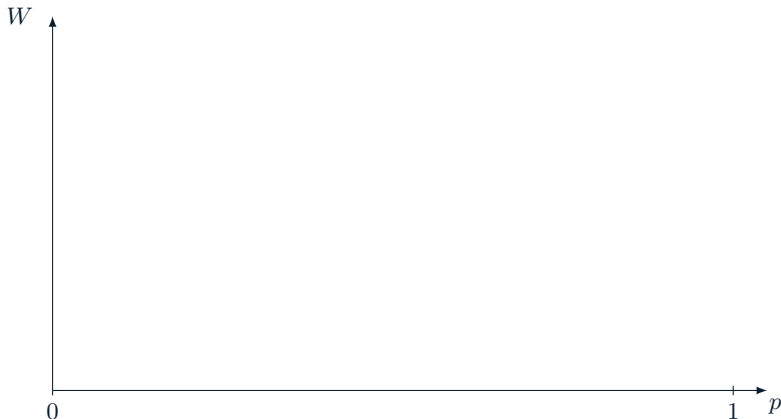
- ▶ Receiver incentive compatibility:

$$\mathbb{E} \left[\underbrace{u_R(p_1^*) - M_1^*}_{\text{Receiver equilibrium payoff}} \mid \mathcal{F}_t \right] \geq \underbrace{u_R(p_t^*) - M_{t-}^*}_{\text{Receiver deviation payoff}}$$

- ▶ If either player deviates from equilibrium, the worst punishment is static autarky.
- ▶ Data (information) sharing equilibrium if we restrict the Sender to data (information) sharing strategies.

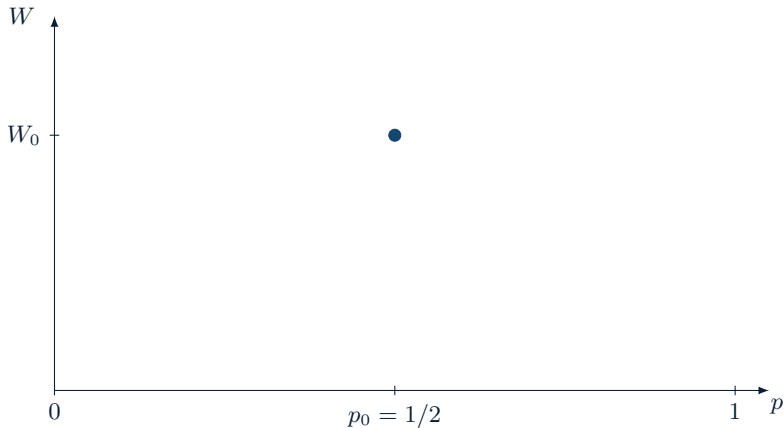
Selling Info without Commitment

Equilibrium Construction/Illustration



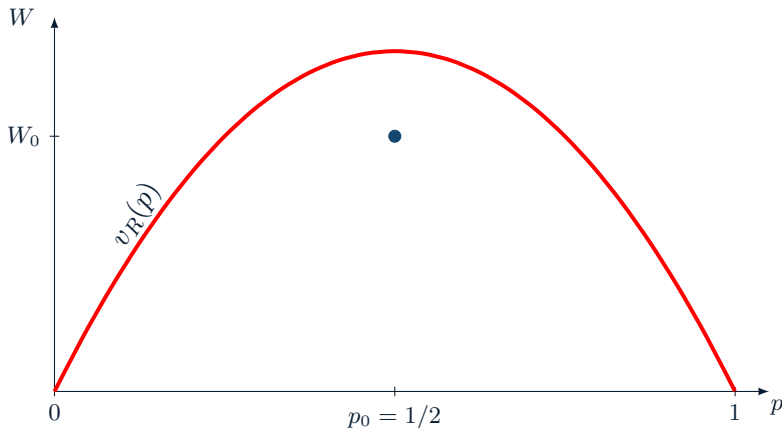
- ▶ Posterior $p_t = \mathbb{E}[\theta \mid \mathcal{F}_t]$ after sharing data up to time t .
- ▶ Expected continuation payment $W_t = \mathbb{E}[M_1 \mid \mathcal{F}_t] - M_{t-}$.

Equilibrium Construction/Illustration



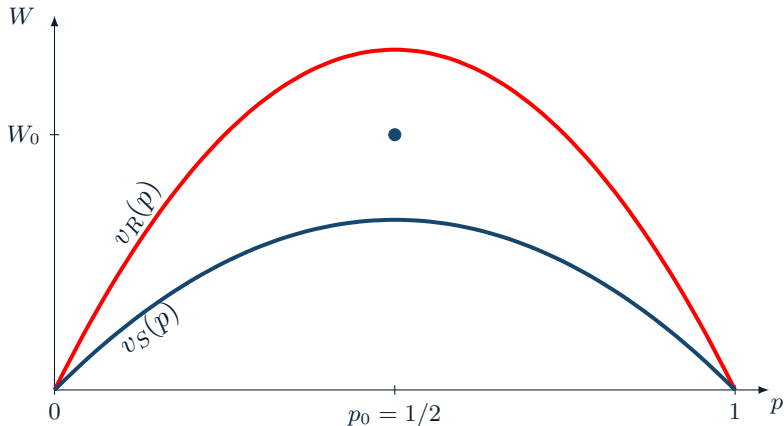
- Start at prior $p_0 = 1/2$ and expected total payment W_0 .

Equilibrium Construction/Illustration



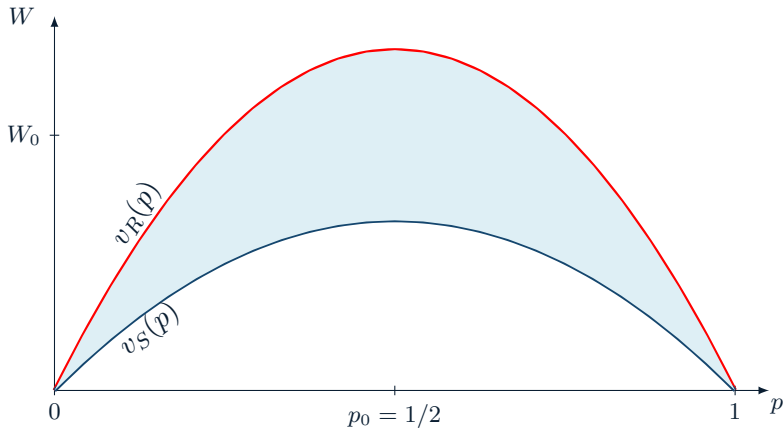
- ▶ Receiver's willingness to pay $v_R(p) = E[u_R(\theta)] - u_R(p)$.
- ▶ Since $v_R(p_0) \geq W_0$, Receiver is ex-ante willing to spend W_0 to learn θ .

Equilibrium Construction/Illustration



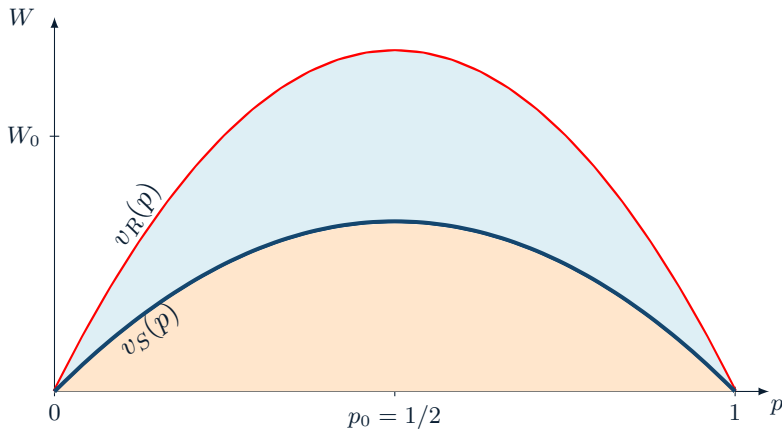
- ▶ Sender's cost of sharing $v_S(p) = u_S(p) - E[u_S(\theta)]$.
- ▶ Since $v_S(p_0) \leq W_0$, Sender is ex-ante willing to share θ in exchange for W_0 .

Equilibrium Construction/Illustration



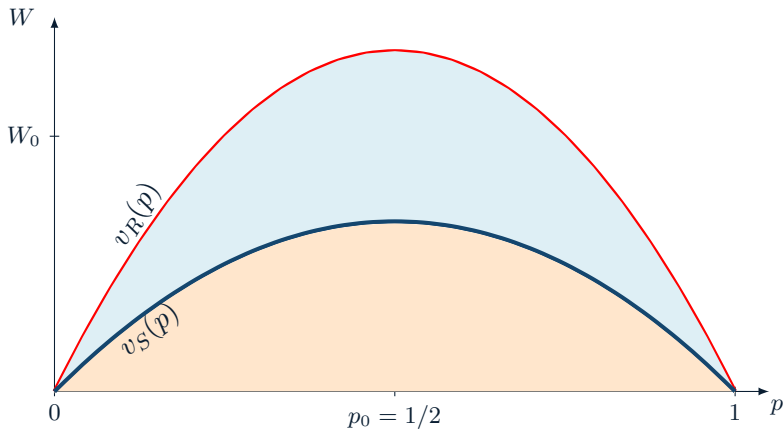
- Ex-ante incentive compatible pairs of (p_0, W_0) at which players are willing to exchange all information about θ for expected payment W_0 .

Equilibrium Construction/Illustration



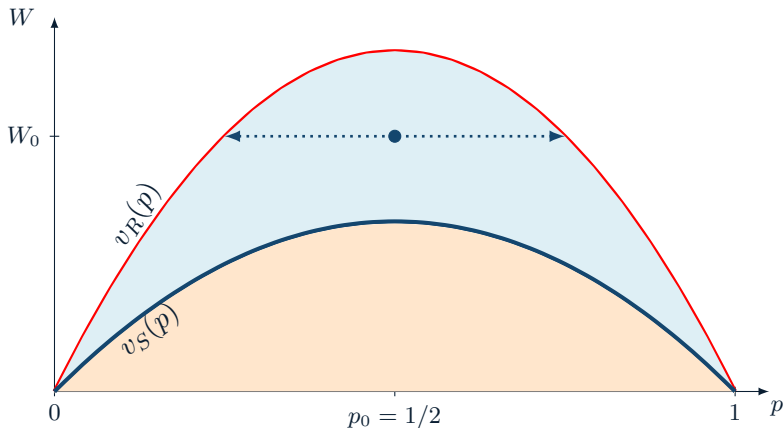
- Below $v_S(p)$, the Sender's continuation cost of information $v_S(p)$ exceeds promised payment W .

Equilibrium Construction/Illustration



- Above $v_R(p)$, the promised payment W exceeds Receiver's benefit from learning information $v_R(p)$.

Equilibrium Construction/Illustration

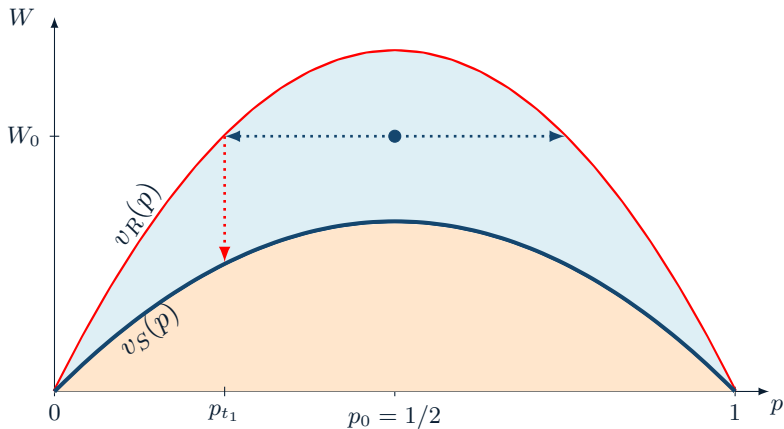


- Equilibrium construction: Sender shares data

$$D_{t_1} \stackrel{def}{=} \inf \{I > 0 : v_R(P(\theta = 1|X_I)) = W_0\}.$$

- The block size D_{t_1} is a first hitting time of $v_R(p_t)$.

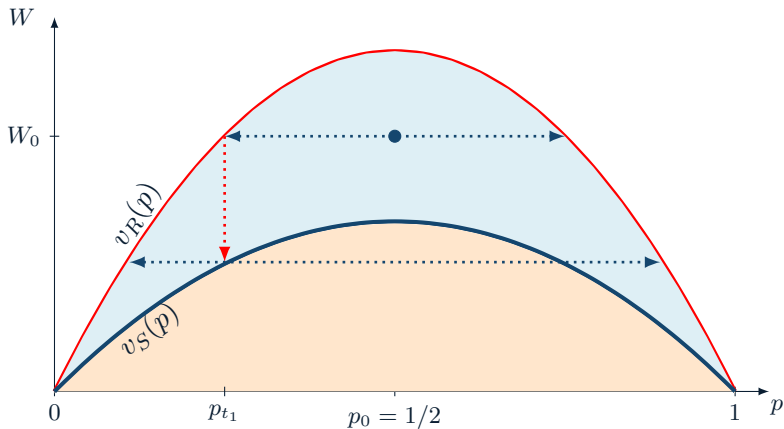
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- Once the Receiver observes the realized belief, he pays the Sender

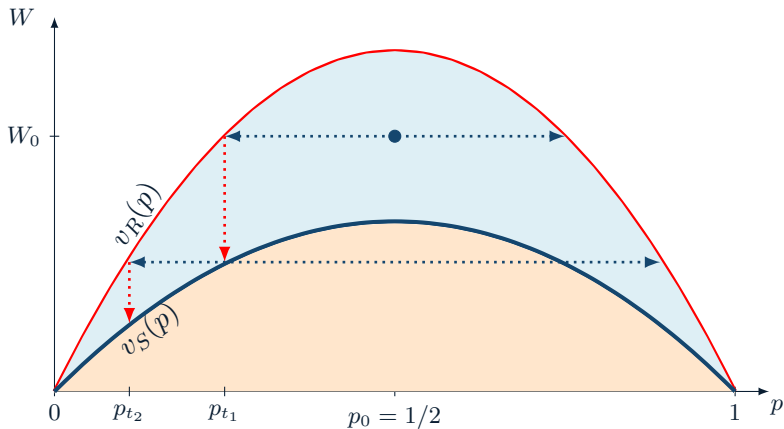
$$M_{t_1} = W_0 - v_S(p_{t_1}).$$

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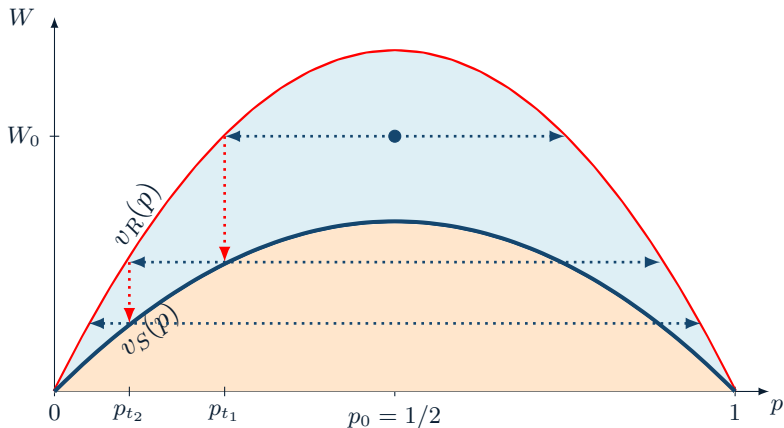
- Players take turns sharing data and making payments.

Equilibrium Construction/Illustration



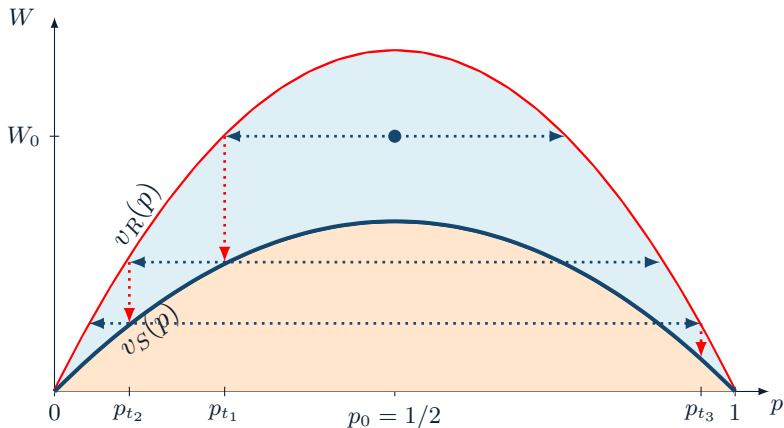
- Every time Sender shares data, he reduces the residual information cost $\Rightarrow$ temptation value declines.

Equilibrium Construction/Illustration



- Every time Receiver makes payment, he reduces his residual budget $\Rightarrow$ his own temptation value declines.

Equilibrium Construction/Illustration



- Sender's residual information cost declines in lockstep with Receiver's residual budget.

Efficiency of Selling Information

Proposition: (Efficient Information Sharing)

There exists an efficient information-sharing equilibrium if and only if the Receiver's budget exceeds the Sender's ex-ante information cost:

$$\bar{M} \geq v_S(p_0).$$

- ▶ Static ex-ante feasibility of full information sharing is sufficient for dynamic implementation.

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 - Each block must be Goldilocks informative ex-post.
 - Otherwise:
 - extremely informative blocks lead to Receiver hold-up;
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- ▶ Information sharing behaves a lot like capital because we are able to generate predictable reductions in residual uncertainty.
- ▶ **Contingent blocks are necessary for efficient communication.**

Selling Data without Commitment

Selling Data: Impossibility

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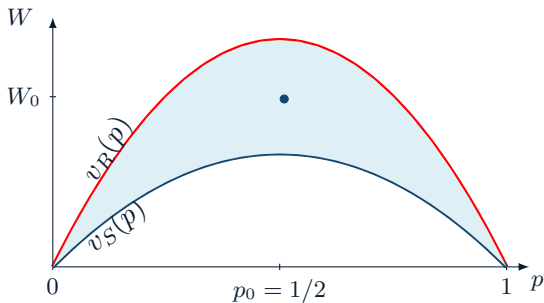
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 - New data can move the prior excessively in any direction

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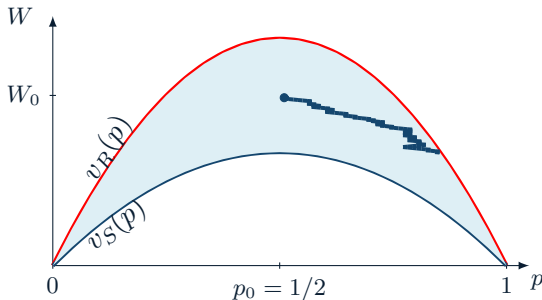


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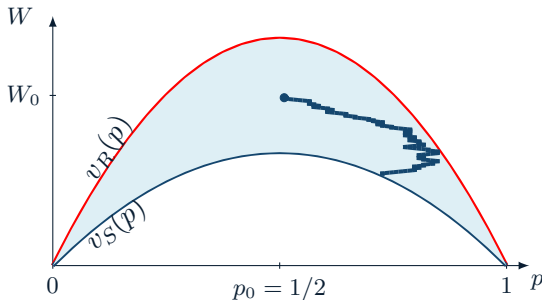
- ▶ Initial data can be too informative $\Rightarrow$ Receiver hold-up.
- ▶ Receiver's payments lower its temptation value and allow the communication to continue.

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- ▶ Subsequent data can be too noisy $\Rightarrow$ Sender hold-up.
- ▶ Sender cannot lower its temptation value by sharing more data, and efficient data sharing breaks down.

Two Simple Extensions

Role of Irreversible Transfers

- ▶ Due to noise, new data may increase uncertainty
 - To incentivize the Sender to share data she needs to get paid
- ▶ With irreversible transfers the Receiver eventually runs out of budget.
- ▶ With reversible transfers Receiver can claw back payments when new data increases uncertainty.

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Lemma:

If transfers are reversible, then efficient data-sharing is feasible if and only if $\bar{M} \geq \max_p v_S(p)$.

- ▶ With high enough budget $\bar{M}$ efficient data sharing is possible.
- ▶ However, ex-ante sufficient budget $\bar{M}$ may be insufficient ex-post.
 - When uncertainty can increase above the initial $t = 0$ level.

Beyond Binary Fundamentals θ

Variance-Based Preferences

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- ▶ Fix variance-based preferences for both players:
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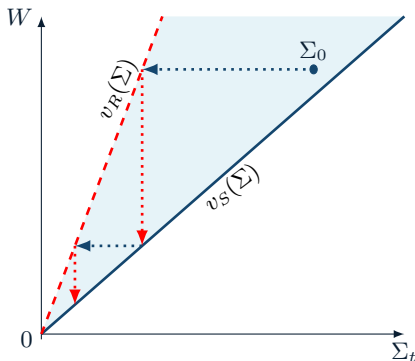
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 - Sender: $u_S(\Sigma)$, increasing in Σ ; $v_S(\Sigma) = u_S(\Sigma) - u_S(0)$;
- ▶ Full information is efficient: $u_R(0) + u_S(0) \geq u_R(\Sigma) + u_S(\Sigma)$ for all Σ .

Efficiency of Information Sharing

Proposition: (Information Sharing Efficiency)

There exists an information sharing equilibrium that attains efficiency whenever the Receiver's starting budget exceeds the Sender's static information cost, i.e., $\bar{M} \geq v_S(\Sigma_0)$.

- ▶ In the limit as all data is shared, posterior variance declines to 0.
- ▶ The size of each block is random, but the posterior variance can be predictable.
- ▶ Consider data batches that reduce posterior variance to pre-specified levels.



$$D_{t_k} = \inf \{ I : v_R(\Sigma(I, X_I)) = \bar{M} - M_{t_k-} \}$$

Normal Priors are Special for Data Sharing

Proposition: (Excess Budget)

Gaussian prior θ is the only family of distributions for which $\bar{M} \geq v_S(\Sigma_0)$ is necessary and sufficient to support an efficient data-sharing equilibrium for any Sender/Receiver preferences that feature gains from trade:
 $v_R(\Sigma) \geq v_S(\Sigma)$ for every $\Sigma \leq \Sigma_0$.

- ▶ If θ is Gaussian, then having just enough budget to compensate the Sender's ex-ante information cost $v_S(\Sigma_0)$ is sufficient.
 - Standard result. Data behaves just like capital.

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 - Standard result. Data behaves just like capital.
- ▶ If θ is non-Gaussian, then efficient data-sharing requires excess budget $\bar{M}$ to absorb dynamic uncertainty.
 - Distributions “closer” to Gaussian require less slack.
 - Distributions similar to binary, e.g. discrete, will require “infinite” slack.

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Gaussian prior θ is the only family of distributions for which $\bar{M} \geq v_S(\Sigma_0)$ is necessary and sufficient to support an efficient data-sharing equilibrium for any Sender/Receiver preferences that feature gains from trade:
 $v_R(\Sigma) \geq v_S(\Sigma)$ for every $\Sigma \leq \Sigma_0$.

- ▶ If θ is Gaussian, then having just enough budget to compensate the Sender's ex-ante information cost $v_S(\Sigma_0)$ is sufficient.
 - Standard result. Data behaves just like capital.
- ▶ If θ is non-Gaussian, then efficient data-sharing requires excess budget $\bar{M}$ to absorb dynamic uncertainty.
 - Distributions “closer” to Gaussian require less slack.
 - Distributions similar to binary, e.g. discrete, will require “infinite” slack.
- ▶ The magnitude of the excess budget depends on the magnitude of gains from information exchange.

Gradualism and Information: Takeaway

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- ▶ Gradual information revelation creates short-term uncertainty risks.
- ▶ Information design effectively hedges such uncertainty risks.
- ▶ Granular data sharing can lead to inefficiencies.
 - Efficiency is impossible for discrete fundamentals.
 - Excess budget is needed for dynamic implementation for all non-Gaussian fundamentals.
 - The magnitude of additional financial flexibility is decreasing in the gains from information trade.

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- ▶ Thank you!